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Orbbec - A

Vision Unlocked: Initiate with OW, the eyes of Embodied AI

We are initiating coverage on Orbbec, a leading 3D vision solution provider, with an Overweight rating and December 2027 price target of Rmb150, implying 38% potential upside (23x FY27E P/S). Orbbec is the platform leader in 3D machine vision, uniquely positioned to capture the next wave of embodied AI and humanoid robotics growth. The company's current c.95% market share in China's humanoid robot vision hardware (per management), full-stack technology platform, and deep integration with major robotics ecosystems create a highly scalable and defensible business model. Orbbec's early-mover advantage in R&D, partnerships with leaders like NVIDIA and Microsoft, and proven ability to deliver plug-and-play vision solutions across all mainstream modalities have established high switching costs and a strong moat. With triple-digit revenue growth in robotics applications, management raising 2026E sales guidance to Rmb1.6–2B, and a rich customer base including Agibot and Unitree, we see Orbbec as a high-conviction play on the scalable, high-growth humanoid robotics supply chain.

- **Platform moat, supply chain strength, and ecosystem integration support premium margins and content value upside.** Orbbec's strategic position as the industry's middleware, combined with its proprietary R&D, in-house chip design, and manufacturing discipline, enables real-time, high-accuracy 3D data output and ensures product reliability. The company's business model is validated by its 2025 turnaround in both profit and Op-CF, and its scalable platform is set to benefit from both domestic and overseas demand as capacity ramps in Vietnam and China. As robotics intelligence and complexity increase, management expects content value per robot to rise, with GPM around 45% and NPM potentially reaching 30–40% in a blue-sky scenario.
- **High market share and pricing power are anchored by technology breadth, ecosystem stickiness, and customer trust.** Orbbec's c.95% share in China's humanoid robot vision market is the result of years of full-stack R&D investment, vertical integration, and a proven track record of delivering reliable, high-performance solutions. The company's technology covers all mainstream 3D vision modalities, allowing it to serve a wide range of OEM needs as the industry evolves. Deep partnerships with ecosystem leaders like NVIDIA, Microsoft, and top domestic robot OEMs create high switching costs and embed Orbbec's solutions in next-generation reference designs. This ecosystem stickiness, combined with a reputation for quality and rapid customization, enables Orbbec to sustain premium ASPs and margins, even as the market scales and new entrants emerge.
- **Our PT highlights Orbbec's high growth potential, but we recognize key risks as the industry matures.** Our valuation reflects the industry's early stage and Orbbec's pivotal role in the humanoid robot value chain. AIoT—including robotics and 3D printing—remains the main growth engine, with consensus FY25-28E sales and earnings CAGRs of c.60% and 90%. High beta and sector volatility, along with the risk of technology shifts away from hardware-centric 3D vision, could challenge Orbbec's market share and pricing power.

Initiation

Overweight

688322.SS, 688322.CH

Price (05 Aug 26): Rmb109.02

Price Target (Dec-27): Rmb150.00

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Sources for: Style Exposure – J.P. Morgan Global Markets Strategy; all other tables are company data and J.P. Morgan estimates.

See page 26 for analyst certification and important disclosures, including non-US analyst disclosures.

Price Performance



	YTD	1m	3m	12m
Abs	21.6%	-26.0%	37.0%	46.0%
Rel	28.9%	-34.2%	40.0%	49.4%

Company Data

Shares O/S (mn)	401
52-week range (Rmb)	153.30-72.45
Market cap (\$ mn)	6,480
Exchange rate	6.75
Free float (%)	44.6%
3M ADV (mn)	20.46
3M ADV (\$ mn)	351.8
Volatility (90 Day)	88
Index	MXCNX
BBG ANR (Buy Hold Sell)	9 0 0

Key Metrics (FYE Dec)

Rmb in millions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	941	1,646	2,684	4,298
Adj. EBIT	113	308	627	1,099
Adj. EBITDA	163	363	693	1,178
Adj. net income	128	315	614	1,037
Adj. EPS	0.32	0.77	1.49	2.52
BBG EPS	0.34	0.77	1.48	2.29
Cashflow from operations	83	277	542	937
FCFF	13	9	122	610
Margins and Growth				
Revenue Growth Y/Y (%)	66.7%	75.0%	63.0%	60.2%
EBIT margin	12.1%	18.7%	23.4%	25.6%
EBIT Growth Y/Y (%)	(211.0%)	171.3%	103.7%	75.3%
EBITDA margin	17.4%	22.0%	25.8%	27.4%
EBITDA Growth Y/Y (%)	(616.3%)	122.3%	91.0%	69.8%
Net margin	13.6%	19.2%	22.9%	24.1%
Adj. EPS growth	(302.8%)	140.5%	94.8%	68.8%
Ratios				
Adj. tax rate	7.4%	3.0%	5.0%	8.0%
Interest cover	NM	NM	NM	NM
Net debt/Equity	NM	NM	NM	NM
Net debt/EBITDA	NM	NM	NM	NM
ROE	4.4%	8.7%	13.4%	19.2%
Valuation				
FCFF yield	0.0%	0.0%	0.3%	1.4%
Dividend yield	-	-	-	-
EV/Revenue	47.1	26.9	16.5	10.3
EV/EBITDA	271.5	122.1	63.9	37.6
Adj. P/E	341.9	142.1	73.0	43.2

Summary Investment Thesis and Valuation

Investment Thesis

We are Overweight on Orbbec, a leading 3D vision solution provider based in Shenzhen, China. Orbbec is strategically positioned as a platform leader in 3D machine vision, a critical enabler for the emerging embodied AI era. The company has an estimated 95% market share (according to management) in China's humanoid robot vision hardware segment, supported by its full-stack technology platform. Orbbec's solutions are embedded in major robotics ecosystems, with partnerships spanning NVIDIA, Microsoft, and leading domestic players.

Valuation

Our December 2027 price target is Rmb 150 based on an FY27E P/E multiple of 23x. We believe this premium is justified by Orbbec's dominant 95% market share in humanoid robot machine vision and its strong exposure to the sector's volume ramp.

Investment Summary

We are initiating coverage of Orbbec, a leading 3D vision solution provider based in Shenzhen, China, with an Overweight rating. Orbbec is uniquely positioned as a platform for 3D machine vision, poised to capture emerging growth opportunities as embodied AI and robotics drive strong demand for spatial information capture. As an early mover, Orbbec benefits from a favorable market landscape, with RealSense (formerly Intel) as the only scalable competitor in this niche, resulting in Orbbec holding c.95% market share in China's humanoid robot applications. Our December 2027 price target is Rmb150, implying 38% potential upside, based on a P/S methodology that reflects the industry's early stage and high growth potential.

Our price target reflects Orbbec's high growth potential, but we recognize the risks of early-stage commercialization and market evolution.

It captures the high growth potential of the embodied AI robotics supply chain, but we acknowledge the risks inherent in a nascent market. Currently, AIoT—including robotics and 3D printing—is the main growth driver for Orbbec, with consensus sales and earnings growth expectations at c.60% and 90% CAGR, respectively, underpinning our FY27E P/S valuation target multiple of 23x. While the outlook is compelling, investors should monitor the pace of commercialization, competitive dynamics, and the sustainability of margin expansion as the market matures.

A proven formula of supply chain strength, full-stack technology, and ecosystem integration underpins Orbbec's market leadership

Orbbec's leadership is built on a strategic supply chain position, a full-stack technology reserve, and an established ecosystem. The company acts as the industry's middleware platform, delivering plug-and-play vision solutions for robotics developers. Deep partnerships with computing platforms such as NVIDIA further reinforce its position. Orbbec's integrated R&D strength—especially in-house chip design and development—enables real-time depth calculation and robust 3D data output, while in-house manufacturing ensures product quality and reliability.

Orbbec's vision hardware is irreplaceable in the humanoid robot supply chain, with reliability and scalability as key differentiators

Vision hardware's unique value lies in its stability and accuracy for real-time data collection, which is critical for model development and deployment as embodied intelligence advances. Robot OEMs typically do not compete in vision hardware, making outsourcing the norm and reliability the primary selection factor. Orbbec's early-mover advantage and proven track record have made it the preferred partner as OEMs scale up.

The company is poised to capture rising content value and margin expansion as robotics intelligence and complexity increase

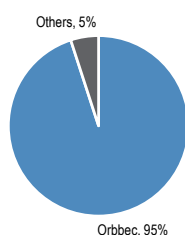
Orbbec stands out among hardware suppliers for its higher potential to increase content value along the supply chain, as the number of 3D vision cameras and sensors per robot is set to rise with greater intelligence and more complex real-world tasks. High barriers to entry support a favorable market landscape and margin profile, with GPM above 50%. The business model is proven, as Orbbec achieved a turnaround in both profit and Op-CF in 2025 during the early stage of robotics market expansion. In a blue-sky scenario, management sees NPM potentially reaching 30–40%, leveraging algorithmic capabilities and a B2B business model.

Table 1: Key forward-looking questions for investors

Question	Why It Matters	What to Watch
Beyond robotics and 3D scanning can Orbbec secure a third scalable end demand market soon?	Diversifying growth pockets to provide long term runway for Orbbec's topline growth	New 3D vision applications, new customer announcements
What could disrupt Orbbec's ecosystem moat?	Ecosystem is a key barrier to entry and margin driver	New partnerships among competitors, customer defections, platform compatibility announcements
How will new entrants impact margin profile?	Margin compression is a key risk to the thesis	Pricing trends, competitor R&D announcements, ASP changes
Will AI model advances reduce hardware differentiation?	Could shift value away from Orbbec's core	Industry adoption of vision-only or monocular solutions

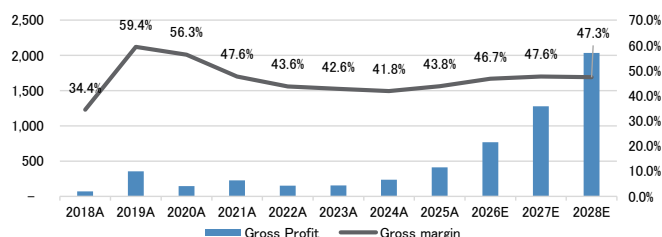
Source: J.P. Morgan.

Figure 1: Key chart 1 - Orbbec's 3D Vision current market position in humanoid/embodied AI Robot applications



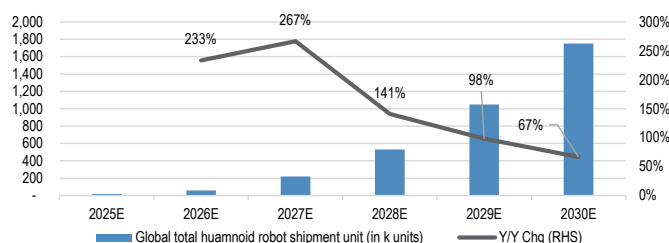
Source: Company data

Figure 3: Key chart 3 - Orbbec's strong margin profile



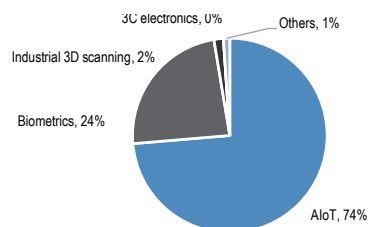
Source: Company reports, J.P. Morgan estimates. Gross Profit in Rmb MM

Figure 2: Key chart 2 - Orbbec to capture the 150% 2025-2030E CAGR opportunity in humanoid/embodied AI robot market



Source: J.P. Morgan estimates

Figure 4: Key chart 4 - AIoT (Robotics and 3D scanning) is Orbbec's key driver (2026E)



Source: J.P. Morgan estimates

Valuation & Share Price Review

Valuation Methodology

We use a price-to-sales (P/S) multiple to value Orbbec, reflecting the early-stage nature of the humanoid robot industry, where many supply chain players remain loss-making due to heavy R&D investment and the nascency of commercialization. Although Orbbec achieved breakeven in FY2025, topline growth remains the most important driver for this growth name, given the sector's rapid expansion and the company's leadership in 3D vision. Our December 2027 price target of Rmb150 corresponds to a FY27E forward P/S multiple of 23x. We believe this premium is justified by Orbbec's dominant 95% market share in humanoid robot machine vision and its strong exposure to the sector's

volume ramp.

Direct comparables are limited due to Orbbec's niche positioning, but we view Leader Drive as the closest A-share peer, given its role as a core component supplier in the humanoid robot value chain, high barriers to entry, and similar margin profile. Both companies are poised to benefit from robust domestic OEM shipment growth. Within the comp table, Hesai and Robosense are also relevant as broad-based machine vision sensor players, but they trade on P/E multiples due to their more mature business mix and LiDAR exposure to the auto market. Newly listed robotics hardware names, such as Laifual Drive and Seer Intelligent, have set IPO valuations at c.20x forward P/S, highlighting strong market appetite for niche, high-growth platforms. We believe Orbbec's valuation should be anchored at the upper end of the comp table, supported by its pivotal role in the humanoid robot value chain, a more defined industry landscape, and additional growth potential from end-demand markets outside robotics, such as 3D scanning.

Table 2: Orbbec comp table

	SBG Ticker	JPM rating	Last Price LC	JPM PT LC	Upside/ Downside	Mkt Cap US\$m	P/E (x)		EPS growth Y/Y		ROE (%)		P/S (x)		Revenue growth Y/Y	
							FY26E	FY27E	FY26E	FY27E	FY26E	FY27E	FY26E	FY27E	FY26E	FY27E
Orbbec	688322 CH	OW	109	150	38%	6,669	142	73	140%	95%	8.7	13.4	27	17	71%	63%
UBTECH	9880 HK	OW	89	161	81%	5,770	NM	189	43%	151%	-5.7	2.9	10	6	100%	59%
Dobot	2432 HK	NC	26	NA	NA	1,305	NM	NM	-25%	58%	-3.9	-0.4	13	9	53%	41%
Robosense	2468 HK	OW	24	45	88%	1,442	53	19	437%	173%	4.6	11.3	3	2	76%	40%
Leader Drive	688017 CH	OW	341	481	41%	9,247	301	197	53%	53%	5.7	8.4	64	45	58%	42%
Hesai	HSAI US	NC	18	NA	NA	2,842	31	20	4%	58%	6.4	9.6	4	3	44%	40%
Serve Robotics	SERV US	NC	6	NA	NA	433	NM	NM	-49%	13%	NA	NA	12	4	NA	201%
Harmonic Drive System	6324 JP	NC	7390	NA	NA	4,514	NM	112	59%	274%	NA	NA	12	10	6%	23%
Horizon Robotics	9860 HK	OW	5	11	105%	10,452	NM	NM	4%	43%	NA	NA	11	7	59%	59%
Average							132	102	74%	102%	3	8	17	11	59%	63%

Source: Bloomberg Finance L.P., as of Aug 05, 2026. J.P. Morgan estimates for covered names, Bloomberg consensus for stocks Not Covered (NC)

Share Price Review

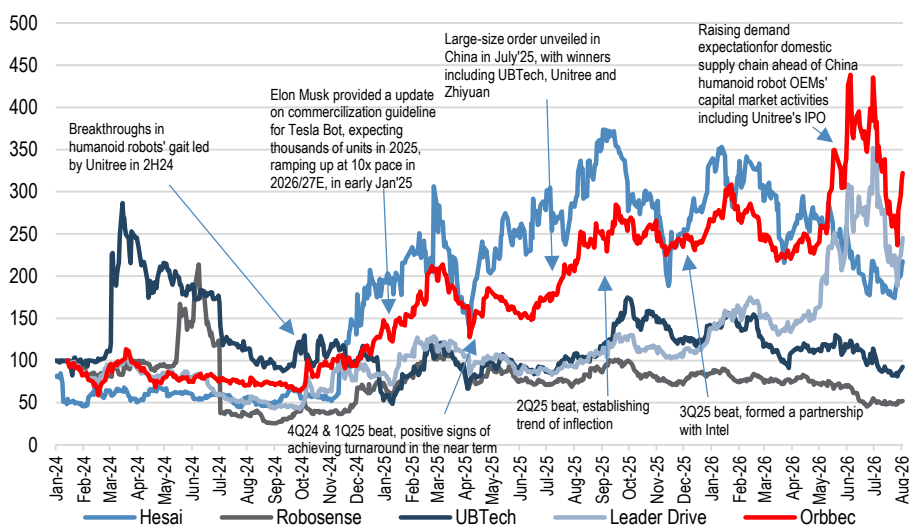
Orbbec's share price has demonstrated high beta and significant volatility since its listing in July 22, rising 252% as of Aug 05, 2026, and outperforming the Star50 Index, which rose 54% over the same period. The stock's performance has been closely tied to sector sentiment, often moving in tandem with other major China humanoid robot supply chain names such as Leader Drive (688187 CH) and UBTECH Robotics (9880 HK). Since the clear inflection point in late 2024—when Orbbec shifted from a technology-intensive R&D phase to a commercial value-capture phase—its share price has exhibited pronounced swings, reflecting both the opportunities and risks associated with the rapid commercialization of humanoid robots.

This elevated volatility is a function of Orbbec's position as a sector proxy and its exposure to industry-wide catalysts. The stock has consistently reacted to news flow around technological breakthroughs, OEM volume ramp plans, and major global events in the humanoid robot space. Notably, Orbbec's share price has rallied sharply on positive sector developments, such as Unitree's advances in gait technology and high-profile announcements from global players like Tesla and Figure AI, but has also experienced pullbacks in periods of uncertainty or when sector sentiment has softened.

We attribute this high beta to Orbbec's advantageous position in the value chain, its niche sub-market with high barriers to entry, and its compelling narrative of rising content value driven by Embodied AI development. The company's attractive margin profile (GPM >40% on a consolidated basis) and strong market share have supported upward earnings revisions, but the stock's trading pattern underscores the importance of monitoring sector catalysts and competitive dynamics. Looking ahead, we expect Orbbec's share price to remain highly sensitive to both domestic and global humanoid

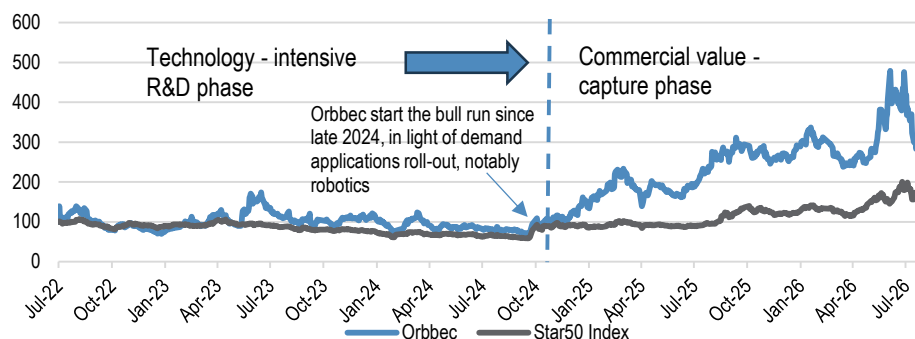
robot supply chain news, with capital activities by domestic OEMs and clear volume ramp-up plans in 2H26 likely to drive further volatility and investor focus on near-term demand visibility.

Figure 5: Key share price events for Orbbec since listing



Source: Bloomberg Finance L.P., Company data, J.P. Morgan. Index against the share price performance of Jan 04, 2024 as 100.

Figure 6: Orbbec's share price performance since listing



Source: Bloomberg Finance L.P., Company data, J.P. Morgan. Index against the share price performance of the listing date Jul 06, 2022, as 100.

Positive Drivers

Positive driver #1: Humanoid robot sector take off is unlocking new opportunities for 3D vision and driving a step-change in Orbbec's revenue potential

The global humanoid robot market is entering a new phase, with China leading the volume ramp and intelligence as the key bottleneck

The past year has seen a dramatic acceleration in the commercialization of humanoid robots, with the "physical body" side—mobility, agility, dexterity, and real-world robustness—making a generational leap forward. Chinese OEMs are now setting the pace globally, moving from lab prototypes and small-batch pilots to scaled deployment

in logistics, manufacturing, and data collection. The latest generation of robots features high DoF, improved stability, and the ability to perform complex, repetitive tasks in harsh environments. Cost-down and scaling are progressing rapidly, with leading Chinese suppliers doubling shipment volumes and achieving price points that are increasingly competitive even by global standards. While the hardware and physical body are advancing at breakneck speed, intelligence—AI models, “brains” and embodied learning—remains the main bottleneck for the sector. The lack of high-quality, real-world data and the challenge of sim-to-real transfer are still limiting the generalization and autonomy of humanoid robots. However, as we have seen positive news from Chinese humanoid/embodied AI robot OEMs claiming that they have realized deployment on edge devices from large vision-language-action (VLA) world models to world action models, we see visible progress towards the “ChatGPT moment” for physical AI. Most importantly for Orbbec, vision is currently one of the most important states in the mainstream brain structure to realize autonomous decision making and action generation, underpinning Orbbec’s strengthening position as a hardware component supplier along the value chain, facilitating humanoid robots to progress into next-level intelligence, rather than a straightforward price performance argument.

Orbbec is poised to benefit from accelerating global shipment growth, rising module intensity, and China’s early mover advantage

Along the path towards fully autonomous humanoid robots or the ‘ChatGPT’ moment, humanoid robot OEMs are making concrete progress in real-world deployment for “low-hanging fruit” tasks—such as sorting and loading in factory and logistics scenarios—while collecting data to enable reinforcement learning and model iteration. This rollout is translating into a visible shipment growth curve and driving down the cost curve, where Orbbec sits as a beneficiary for the tangible volume growth, while we see limited pricing pressure. Our global humanoid robot shipment forecast for 2025–2030E projects a c.150% shipment CAGR as our base-case scenario, reaching c.1.75MM units by 2030E, with China contributing c.50%. China is the early mover, accounting for >80% of global shipments in 2025, driven by manufacturing supply chain readiness and a government-led push to make humanoid robots a showcase for next-generation high-tech. Our near-term forecast (2026–2027E) is based on bottom-up channel checks with OEMs, component suppliers, and distributors, while our long-term view incorporates labor shortage as a demand anchor, adoption curve benchmarks from NEVs and smartphones, and the pace of physical intelligence development. The rest of the world is poised to catch up from 2027E onward, with a c.210% shipment CAGR as global supply chains become more established and physical AI technology matures. We expect Orbbec’s humanoid robot applications to achieve a 35% CAGR in sales terms from 2025 to 2030E, with the share of revenue from humanoid robot applications rising to 54% by 2028E, up from c.5% in 2025E.

Table 3: Implied % revenue upside for Orbbec from our humanoid robot shipment forecasts 2025-2030E

	2025E	2026E	2027E	2028E	2029E	2030E
China market shipment in units	15,000	50,000	120,000	280,000	550,000	900,000
% of solution penetration	65%	85%	90%	93%	95%	97%
% market share	95%	95%	93%	85%	80%	75%
ex-China market shipment in units	3,000	10,000	100,000	250,000	500,000	850,000
% of solution penetration	95%	98%	98%	98%	98%	98%
% market share	5%	40%	40%	40%	40%	40%
ASP-China	5,300	6,320	6,194	5,814	5,506	5,297
ASP-overseas	8,000	8,000	9,500	10,260	11,012	11,579
Orbbec's revenue in Rmb MM (China)	49	255	622	1,287	2,302	3,468
Orbbec's revenue in Rmb MM (Overseas)	1	31	372	1,005	2,158	3,858
Total humanoid robot application revenue in Rmb MM	50	287	995	2,292	4,460	7,326
Y/Y Chg		420%	144%	107%	79%	51%
Contribution from humanoid	5%	17%	37%	53%	67%	74%

Source: Company data, J.P. Morgan estimates.

Table 4: Global humanoid robot shipment forecasts

000 units

	2025E	2026E	2027E	2028E	2029E	2030E
China	15	50	120	280	550	900
ex-China	3	10	100	250	500	850
Global total humanoid robot shipment unit (in k units)	18	60	220	530	1,050	1,750
Y/Y Chg (RHS)		233%	267%	141%	98%	67%
Global humanoid robot TAM in Bn units according to Elon Musk estimate	5	5	5	5	5	5
Global humanoid robot penetration rate	0.00%	0.00%	0.00%	0.01%	0.02%	0.04%

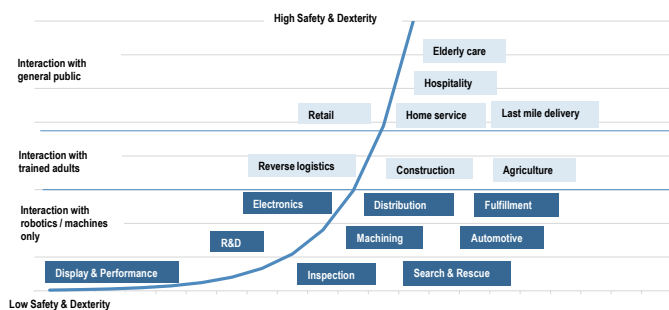
Source: J.P. Morgan estimates

Table 5: Our key forecast assumptions for humanoid robot shipment forecasts

Physical AI development stage	Our definition	Penetration rate upside for different scenarios			
		Industrial	Retail and warehouse	Other commercial	Household
S1	Satisfactory accuracy and productivity for single task performance in structured environment	5%	1%	2%	0%
S2	Satisfactory accuracy and productivity for a few tasks' performances (but still short-horizon tasks), more adaptive to changing environment	35%	20%	2%	0%
S3	Realizing single scenarios' 0-shot generation (across tasks)	65%	65%	15%	0%
S4	Realizing cross scenario, cross task 0-shot generation, with minor model fine-tuning needed	85%	90%	50%	2%
S5	Fully 0-shot generation, similar to a human being's capability	95%	98%	80%	15%

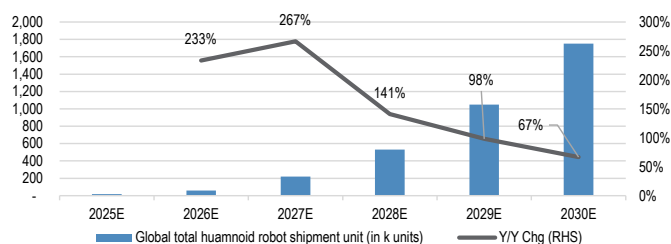
Source: J.P. Morgan.

Figure 7: Global humanoid robot user-case expansion roadmap



Source: J.P. Morgan estimates

Figure 8: Global humanoid robot shipment forecast in units



Source: J.P. Morgan estimates

Positive driver #2: Orbbec is leveraging its leadership position in 3D vision to capture design wins and sustain premium content value

Orbbec's platform approach, technology breadth, and early-mover advantage are supporting design wins across a fragmented and evolving market

Orbbec adopts a platform approach, focusing on the B2B market as a “tool seller,” providing standardized 3D vision sensors for all types of robotics—including humanoid robots, service robots, and industrial robots. Orbbec is one of the very few sensor technology platforms that covers all mainstream 3D vision technology roadmaps. Its core competitiveness lies in its hardware platform, which supports a full range of technologies, including structured light, stereo vision, and ToF. In addition, the company has vertically integrated R&D capabilities, most notably in-house chip design and development, which enable Orbbec's algorithms to perform real-time depth calculation and output 3D data—a core aspect of 3D vision technology. The industry has not yet reached a consensus on the optimal 3D vision technology roadmap, and every choice involves trade-offs, balancing performance, cost, power consumption, form factor, and algorithmic complexity. Orbbec's full-stack technology reserve and early-mover advantage position it as a one-stop-shop 3D vision platform, well placed to benefit from the take-off of the humanoid robot sector.

A rich customer base, ecosystem integration, and robust product matrix support Orbbec's content value and margin profile

Currently, Orbbec's humanoid robot machine vision shipments are mainly binocular stereo vision with structured light, which generated c.Rmb50MM in orders from humanoid robot applications in 2025. Beyond structured light, Orbbec also offers solutions including passive stereo vision and iToF, covering nearly all mainstream 3D vision technologies adopted in the market. Orbbec holds a leadership position in China, with an estimated 95% market share in the domestic humanoid robot market. Its customers include Agibot, Unitree, UBTech, Beijing HRIC, Booster Robotics, Astribot, and others, with Agibot, Unitree, and UBTech together accounting for 78% of total humanoid robot shipments in China in 2025. Orbbec's 3D vision solutions are deeply embedded in mainstream computing platforms, including NVIDIA Isaac, Microsoft, Intel, and OpenHarmony, enabling full platform compatibility and plug-and-play functionality for developers. Management assumes an ASP of c.Rmb2,000 per module, and as applications diversify, the number of modules per robot is expected to increase from 1–2 units to at least 4, and potentially reach 6–10 units for more advanced interaction requirements. This supports a robust content value per robot and a GPM for humanoid robot applications likely to remain above 50%, well above other key hardware components.

Table 6: Summary of 3D solutions fitting into different robotic functional needs

Need	Best output type	Common sensor source
Semantic understanding	RGB image, segmentation, object detections	Monocular or binocular RGB cameras
Metric depth	Depth map, RGB-D image	iToF, structured light, stereo
3D world modeling	Point cloud, occupancy map, mesh	dToF, stereo, structured light
Robot manipulation	Dense close-range depth, RGB-D, point cloud	Structured light or iToF
Navigation/mapping	Point cloud, range image, occupancy grid	dToF or stereo

Source: J.P. Morgan

Positive driver #3: Orbbec's exposure to humanoid robots is creating a sweet spot within the value chain and supporting content value upside

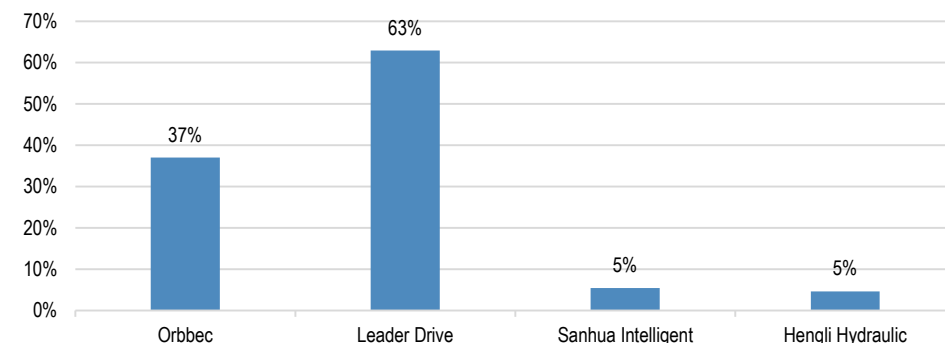
Orbbec's sales exposure to humanoid robots is one of the largest in the supply chain, making it a strong proxy for sector growth

In 1Q26, Orbbec's sales from humanoid robot applications tripled in China, reaching c.20% of total sales. We expect this contribution to increase further, reaching 37% by 2027E. Among our coverage of the humanoid robot supply chain, Orbbec's sales exposure to humanoid robots is one of the largest, alongside Leader Drive. This is due to the niche nature of its existing businesses and its strong market position within the domestic supply chain, which leads the pace of commercialization. As a result, Orbbec serves as a strong proxy for capturing the exponential growth that we expect in the domestic humanoid robot supply chain.

Content value per robot is poised to rise as module count increases and vision becomes a larger share of the BOM

According to management, the average content value is c.Rmb2,000 per module unit, depending on the specific vision solution. Management anticipates that content value will increase over the next 3–5 years, as the number of modules required per humanoid robot rises. As humanoid robots transition from teleoperation-controlled stages to more advanced intelligence levels, autonomous robots will require more multimodal data collection and faster vision data processing for real-world deployment. This means the number of modules could increase from 1–2 units to at least 4, with some humanoid robots in the market already featuring up to 10 module units. Visual and sensor components represent one of the largest shares in a humanoid robot's bill of materials (BOM), at around 20%, while 3D vision specifically accounts for a smaller portion, with content value estimated at c.Rmb3,000–5,000. Nevertheless, 3D vision is one of the few supply chain components with significant potential for content value upside and BOM share expansion as physical AI technology advances.

Table 7: Differences in % sales contribution from humanoid robot in 2027E across select Chinese component plays



Source: J.P. Morgan estimates

Table 8: Humanoid robot BOM estimate

Categories	% of BOM estimate	Note
Reducer and roller screw	33%	reducer and roller screw would be used in rotary and linear actuator structure respectively
Motor and motor drive	15%	motor and motor drive are widely used in different actuators to realize degree of freedom including joints and hands
Sensor and visual	20%	machine vision, pressure sensor, IMU, encoder, GPS and LIDAR
Controller	5%	CPU, GPU, chips
Enclosure, material	12%	
Cables, connectors and mechanical parts	10%	
Battery and power management	5%	battery pack, battery management system, and thermal management system

Source: J.P. Morgan estimates. Note that we do not include dexterous hands, as the pricing varies hugely depending on DoF (degree of freedom), and it is optional for some humanoid robot shipments at present. According to the supply chain, it should typically account for 15-30% of total BOM if included. The high end dexterous hands' pricing is close to US\$50k per unit at present, exceeding a low end humanoid robot price

Positive driver #4: Orbbec's integrated tech platform and supply-side expansion are supporting robust growth beyond humanoid robots

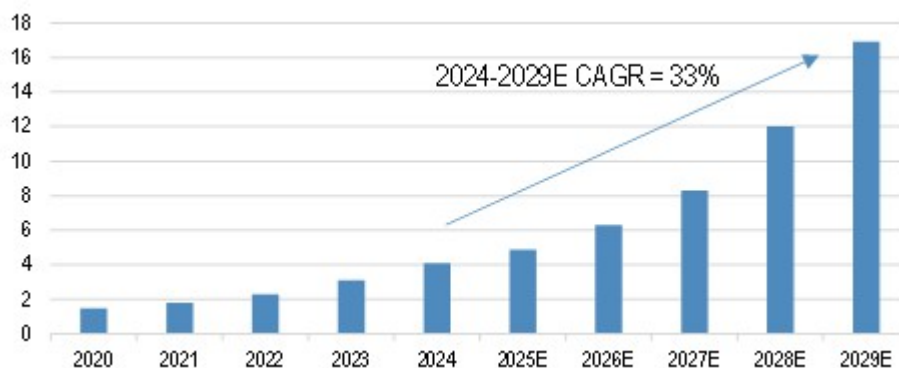
Management is raising sales guidance as supply-side expansion in China and overseas unlocks new growth opportunities

Management has recently raised its 2026E sales guidance to Rmb1.6–2B, representing c.90% Y/Y growth at the midpoint, up from the previous guidance of Rmb1.5–1.6B provided during the 1Q26 briefing. This upward revision reflects faster-than-expected progress in overseas capacity expansion, with the Vietnam factory expected to begin production as early as June or July, boosting overseas sales in the second half of 2026. Looking ahead, Orbbec is planning for overseas capacity with a designed annual output of Rmb4B starting from 2027E. For 2026E, the overseas capacity is expected to deliver a total output value of Rmb1B as the first step per management. This capacity is intended to supply US tech companies, leveraging strong collaborations such as with NVIDIA. Within China, Orbbec is also expanding its capacity, aiming to support annual sales of Rmb6B once the designed capacity is reached.

AIoT and new use cases are providing additional growth engines, with strong demand outlook beyond robotics

Currently, AIoT is the key growth driver for Orbbec, accounting for 54% of total sales in 2025. Within this segment, AIoT can be further broken down into robotics and 3D scanning applications, which together make up c.90% of total AIoT sales in 2025. Orbbec recorded c.Rmb300MM in sales from broad-based robotics applications in 2025, which saw triple-digit growth in 1Q26 and is poised to sustain at least double Y/Y growth for the remainder of the year. For 3D scanning applications, end-demand use cases include consumer (2C) applications such as 3D printing. According to the prospectus of Creality, a key customer of Orbbec, the global consumer 3D printing market achieved a 36% CAGR from 2021 to 2025 and is expected to maintain a 33% CAGR from 2026 to 2030, driven by advances in AI and breakthroughs in material science that facilitate greater market penetration. As AI technology continues to develop, new use cases for 3D vision are emerging, such as object recognition and surveillance, AR/VR gaming, plastic surgery, logistics, and livestock husbandry.

Figure 9: Market size of global consumer 3D printing industry, in terms of GMV, in US B



Source: CIC

Risk Factors

Risk factor #1: Competitive intensity and new entrants could erode Orbbec's market share and pricing power in humanoid 3D vision

The competitive landscape for humanoid 3D vision is evolving rapidly, with new entrants and adjacent technology players increasing the risk of share loss and ASP pressure

Orbbec is currently the only listed pure-play company for 3D vision, equipped with large-scale supply to meet the exponential demand in applications such as robotics and 3D scanning. However, the attractive long-term outlook for humanoid robots and physical intelligence, as well as the high margin profile (GPM >50% for the AIoT segment) is drawing new entrants into the market. RoboSense, for example, has released its AC2 sensor system, integrating dToF LiDAR, a binocular RGB camera, and an IMU sensor, directly targeting humanoid robot applications. LiDAR-focused companies have the funding and mass-production experience to accelerate their penetration, especially as pricing pressure in automotive hardware pushes them to diversify. Orbbec's cumulative R&D expenses totaled Rmb2.2B from 2018 to 2025, but competitors like Hesai and RoboSense are now spending Rmb600–800MM per year, indicating that the competitive landscape could shift quickly if these companies pivot more aggressively toward robotics vision.

Management's key thresholds for risk are market share and GPM, with early warning signs to monitor as the market matures

The thesis would be challenged if Orbbec's market share in China's humanoid robot applications falls below 80% within three years in our view, or if GPM falls below 40% due to pricing pressure or technological disruption. The key watch items are competitor design wins, ASP trends, and any early signs of customer concentration tightening the negotiating dynamic. A shift in bargaining power or a step-down in ASP/module would be a visible early warning signal for investors. The risk is not just about new entrants, but also about the pace at which established players can scale up R&D and manufacturing to match Orbbec's current leadership, potentially compressing both share and margin before the market fully scales.

Risk factor #2: Technology shifts toward software vision and alternative sensing could cap module intensity and content value

Advances in AI models and software vision could reduce the need for depth hardware, capping the module count upside that underpins Orbbec's content-value flywheel

The industry has not yet reached a consensus on the balance between hardware and software vision solutions. While depth cameras provide accurate measurement and robust perception as a physical layer, this comes at the cost of higher hardware expenses and a more cumbersome robot body design. In theory, if end-to-end models are pre-trained on massive RGB datasets and can estimate spatial information effectively, reliance on hardware certainty could limit the emergent capabilities of algorithms. Some robot OEMs, such as Sudo in China, have already delivered impressive demonstrations likely using only monocular cameras, and Tesla is widely recognized for adopting a passive vision solution. The addition of other sensing modalities, such as tactile or force sensors, could further dilute the dominance of vision in the overall system architecture.

Real-world engineering constraints still favor multi-sensor fusion, but the risk of structural module reduction remains as the industry evolves

While real-time processing and power constraints still favor multi-sensor fusion in many scenarios, the risk remains that future reference designs could structurally reduce depth module count, directly impacting the revenue bridge even if overall humanoid shipments continue to grow. The practical risk is not that depth disappears, but that depth intensity declines in certain deployments, which would hit the revenue bridge via modules/robot and mix. The key watch item is whether reference designs structurally reduce depth module count across successive generations. Ultimately, engineers must find the right compromise for each specific model and scenario, balancing power and hardware budgets with the intelligence of their models. As physical AI's development roadmap has not converged, similar concerns also include the addition of tactile sense, force sense, and other state elements which may dilute the dominance of vision.

Risk factor #3: Short listing history, high growth expectations, and execution risk increase the risk of de-rating if delivery wobbles

Orbbec's short listing history and limited valuation record increase the risk of de-rating if growth delivery wobbles or market expectations reset

Orbbec has a relatively short listing history, having only been public since mid-2022. More importantly, the market's total addressable market did not expand significantly

until AI applications became more commercialized and humanoid robot applications entered the spotlight. The company is currently trading at a high P/S and P/E, supported by strong growth expectations for both sales and profit—c.60% and 90% FY2025–2028E CAGRs, respectively per Bloomberg consensus. If Orbbec fails to meet these expectations, its valuation could suffer from a de-rating, accompanied by downward earnings revisions. The short trading history means there is limited track record for reference, and there are no direct comparables for Orbbec’s specific niche market, making it challenging to construct a stable valuation band.

Execution risk around capacity ramp, quality control, and working capital discipline is heightened as the company scales and diversifies

The company only became profitable in FY2025, so the sustainability of profitability and Op-CF must be monitored as user-case expansion and sales pick up. Similarly, Orbbec achieved net operating cash inflow in FY2025 for the first time after five consecutive years of outflows, and it remains to be seen whether this turnaround will be sustained as user-case expansion and sales pick up. Execution risk around overseas capacity ramp, quality control, and working capital discipline also warrants close attention as the company scales. Any deterioration in working capital discipline, ramp timing, or margin delivery would be a visible early warning signal for investors and could trigger a negative re-rating. The company’s DCF and P/S-based valuation frameworks rely on continued high growth and margin delivery, so any miss on these fronts could result in a sharp correction, given the lack of direct comparables and the market’s sensitivity to growth delivery in early-stage, high-multiple names.

Company Background

Orbbec’s evolution from niche 3D vision developer to platform leader in robotics and AIoT is underpinned by vertical integration and technology breadth

Founded in 2013, Orbbec has rapidly transformed from a niche 3D vision sensor developer into China’s dominant platform for machine vision in robotics and AIoT. The company’s trajectory reflects a deliberate strategy of vertical integration and technology breadth, positioning it as a critical enabler in the humanoid robotics supply chain. Orbbec’s early commitment to in-house R&D—evidenced by cumulative R&D spend exceeding Rmb2.1B from 2019 to 2025—has yielded a proprietary portfolio spanning structured light, iToF, stereo vision, and dToF. This full-stack capability is rare among global peers and underpins Orbbec’s c.95% market share in China’s humanoid robot vision hardware. The company’s ability to industrialize new sensor technologies every 2–3 years has allowed it to stay ahead of both domestic and international competitors, many of whom remain focused on single-technology or legacy industrial applications.

Orbbec’s platform model and ecosystem integration create high switching costs and a defensible moat in a rapidly evolving industry

Unlike most sensor suppliers, Orbbec has embedded itself as a middleware platform within major robotics and AI ecosystems. Its solutions go beyond hardware, offering SDKs and developer tools that accelerate customer adoption and stickiness. Partnerships with NVIDIA, Microsoft, and leading Chinese OEMs (e.g., Unitree, UBTECH) provide both technical validation and privileged access to next-generation reference designs. This ecosystem integration creates switching costs for customers and raises barriers to entry for new competitors. Orbbec’s manufacturing model blends in-house production for core modules (e.g., laser projection, imaging, calibration) with outsourced assembly for non-core components, leveraging the Pearl River Delta’s electronics supply chain.

The recent expansion of capacity in Vietnam and China is not only a response to surging demand but also a strategic hedge against geopolitical and supply chain risks. The company's ability to reduce its cash conversion cycle to 27 days in 2025 and achieve positive Op-CF for the first time in 2025 signals operational maturity and scalability.

A broad and diversified customer base supports Orbbec's growth, but concentration risk remains as the market consolidates

Orbbec's customer base is both broad and deep, with over 1,000 clients across biometrics, robotics, 3D scanning, and industrial metrology. The company's future growth is increasingly levered to AIoT and robotics, which we expect to account for c.90% of revenue by 2028E (vs. c.13% in 2019). Key customers such as Ant Group (biometrics), Creality (3D printing), and leading robotics OEMs provide both volume and validation, but also expose Orbbec to customer concentration risk as the market consolidates. Since mass production began at the end of 2015, Orbbec has served more than 1,000 customers worldwide, including industry leaders such as OPPO, Ant Group, Jabil, Muyuan, China Mobile, Matterport, and Beike Rushi.

Orbbec's vertically integrated technology platform and proprietary chip development support rapid innovation and margin resilience

Orbbec's platform covers all mainstream 3D vision modalities—structured light, iToF, dToF, stereo vision, LiDAR, and industrial metrology—enabling rapid adaptation to shifting industry standards and customer requirements. This “full-stack” approach insulates Orbbec from technology obsolescence and allows it to capture a larger share of the bill of materials as robotics and AIoT applications proliferate. Heavy investment in proprietary depth-engine chips and advanced algorithms reduces reliance on external suppliers and secures higher margins through intellectual property ownership. The integration of hardware and software enables real-time, high-accuracy 3D data output, which is critical for robotics applications where latency and reliability are non-negotiable. This vertical integration also accelerates time-to-market for new products and allows for rapid customization to meet the needs of leading OEMs.

Manufacturing integration, quality control, and supply chain management are key enablers of Orbbec's scalability and cost discipline

Orbbec's manufacturing model strategically balances in-house production for core, IP-sensitive modules (e.g., laser projection, calibration, RGB/IR imaging) with outsourced assembly for non-core components. This approach leverages China's mature electronics supply chain while maintaining control over quality and innovation in critical areas. The company's dual-site manufacturing footprint—anchored in the Pearl River Delta and recently expanded to Vietnam—provides both scale and resilience. The Vietnam facility is a strategic hedge against geopolitical and supply chain disruptions, supporting overseas customer growth and reducing single-region risk. In-house assembly is reserved for high-precision steps such as laser module processing, RGB/IR imaging, calibration, and final system integration, ensuring stringent optical alignment and performance standards for leading robotics and AIoT customers.

Orbbec's sales model and management team support rapid customer adoption and long-term innovation

Orbbec primarily adopts a direct-sales model, providing standard 3D vision sensors, an application computing platform, and SDK development tools to accelerate customers' learning and implementation process. For more mature customers, Orbbec offers dedicated technical support and in-depth product customization services, often co-developing new products for potential application markets. The senior management

team has been very stable, with three core technical personnel still serving within the company. The three technology team leaders all hold strong academic backgrounds and are in their 40s, with founders Mr. Huang Yuanhao and Mr. Xiao Zhenzhong together holding c.30% shareholding. Since listing, Orbbec has carried out two rounds of incentive plans, including 219 employees, with KPIs tied to sales or gross profit for 2025–2027E, and has already beaten its own targets one year in advance based on FY2025 earnings delivery.

Industry market size and investor implications highlight both opportunity and the need for continued execution

According to Yole Intelligence, the global 3D imaging and sensing market is poised to reach US\$17.2B by 2028E, implying a 13% 2022–2028E CAGR, with the automotive market expected to be the growth driver at 38.5% CAGR due to ADAS LiDAR and in-cabin sensing. This forecast was made before the emergence of physical AI, and we think robotics is poised to provide further upside. Orbbec's vertically integrated platform is a structural moat that supports premium margins, rapid innovation, and ecosystem lock-in. However, sustaining this advantage will require continued investment in R&D, vigilance against technology commoditization, and proactive management of supply chain and geopolitical risks. Investors should monitor the cadence of new product launches, the evolution of gross margins, and the depth of integration with key ecosystem partners as leading indicators of platform durability.

Table 9: Orbbec's core technology personnel

Core technical personnel	Age	Education / Major	Tenure & position	Technical background
Huang Yuanhao (黄源浩)	46	Major in optical measurement; B.S. Peking University; M.S. National University of Singapore; Ph.D. City University of Hong Kong; postdoctoral research at The Hong Kong Polytechnic University, Ryerson University (Canada), The Chinese University of Hong Kong, and the MIT SMART Center	Since 2013; company founder; currently Chairman & General Manager	Internationally recognized optical measurement expert, led the definition of the company's technology roadmap; published 20+ papers (incl. Optics Letters); served as conference session chair and academic committee member; reviewer for 10+ international journals; as a principal inventor: 359 patent applications, 142 granted patents
Xiao Zhenzhong (肖振中)	46	Major in machine vision & 3D sensing; B.S./M.S./Ph.D. Xi'an Jiaotong University; postdoctoral research at Nanyang Technological University (Singapore); appointed (Nov 2011) as a doctoral supervisor/instructor at Xi'an Jiaotong University, School of Mechanical Engineering	Since 2013; co-founder; currently Director & CTO	During Ph.D., participated in one national "863 Program" project (as team lead) and one National Natural Science Foundation project; contributed to drafting a national standard; published 10+ academic papers; since joining, co-led definition of the company's technology roadmap and organized development of core foundational technologies—incl. leading development (2014) of the company's first "depth engine" algorithm; as core technical staff, participated in ~10 national/provincial/municipal R&D projects; as a principal inventor: 328 patent applications, 129 granted patents
Mei Xiaolu (梅小露)	47	Computer science; B.S. Peking University; Chinese Academy of Sciences, M.S. Institute of Computing Technology	Since Jun 2014; currently Senior Vice President; Head of chip R&D	15+ years of chip design & development experience; led the company's 3D vision sensing chip R&D, delivered the company's first generation 3D vision sensing chips; as a principal inventor: 20 patent applications, 7 granted; as a core member, participated in 3 projects including national key R&D, provincial major special projects, and municipal government S&T projects

Source: Company report

Table 10: KPI to unlock incentive plan

Vesting period	Assessment	Assessment metric (either Revenue (A) or Gross Profit (B)) — Blue-sky target value		Assessment metric (either Revenue (A) or Gross Profit (B)) — Base-case target value	
	year				
1st vesting period	2025	Revenue = Rmb701MM or Gross Profit = Rmb250MM		Revenue = Rmb631MM or Gross Profit = Rmb230MM	
2nd vesting period	2026	Revenue = Rmb900MM or Gross Profit = Rmb330MM		Revenue = Rmb810MM or Gross Profit = Rmb300MM	
3rd vesting period	2027	Revenue = Rmb1100MM or Gross Profit = Rmb400MM		Revenue = Rmb990MM or Gross Profit = Rmb360MM	

Source: Company data

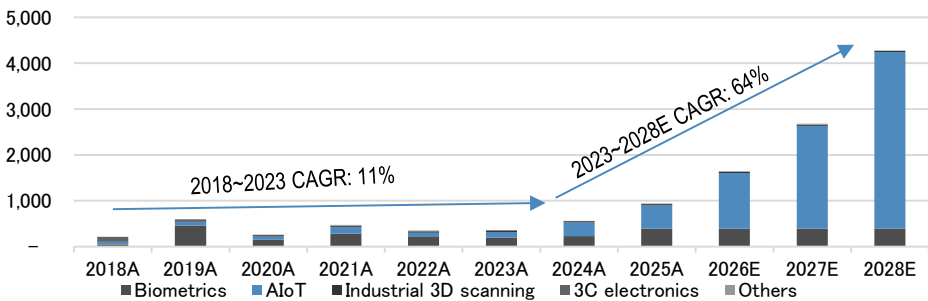
Financial Forecasts

Income Statement

Orbbec’s revenue growth is accelerating as robotics and new applications drive a sharp mix shift toward AIoT.

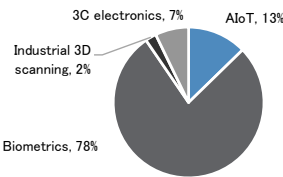
Orbbec delivered a 24% revenue CAGR from 2018 to 2025, reaching Rmb941MM by 2025. Historically, 3D vision was in the early commercialization stage until 2024, but the pace of sales growth has visibly accelerated since then, driven by market TAM expansion in emerging applications—most notably robotics and 3D scanning. AIoT’s revenue contribution jumped from c.30% in 2020–2022 to >50% in 2025, and we expect this ratio to further increase to c.90% by 2028E. We forecast revenue growth of 75%/63%/60% Y/Y in 2026/27/28E, with the AIoT segment as the key growth driver at a 96% CAGR over the forecast period. Humanoid robot exposure is poised to reach 53% by 2028E, up from single digits in 2025, while other AIoT sales are set to deliver a 50% CAGR, led by >30% growth in 3D scanning, new customer penetration, and overseas ramp-up. On the conventional business front, we assume a stable trend for biometrics, with 0% Y/Y growth in 2026–2028E, as domestic payment-related business (e.g., Alipay’s Touch and Pay) has reached a steady state and future growth will be bridged by overseas opportunities.

Table 11: Revenue by segment (2018 ~ 2028E in Rmb MM)



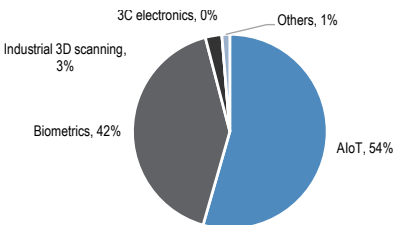
Source: Company data, J.P. Morgan estimates

Figure 10: Revenue breakdown by vertical (2019A)



Source: Company data

Figure 11: Revenue breakdown by vertical (2025A)



Source: Company data (

Table 12: Revenue forecasts by segment (2026~2028E in RmbMM)

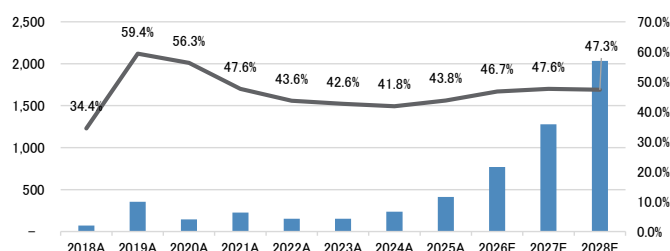
Rmb MMs (PRC GAAP)	2023A	2024A	2025A	2026E	2027E	2028E
Revenue	360	564	941	1,646	2,684	4,298
AIoT	123	298	512	1,213	2,245	3,855
Biometrics	196	231	390	390	390	390
Industrial 3D scanning	31	26	26	26	26	26
3C electronics	1	-	-	-	-	-
Others	10	9	12	17	23	27
Y/Y growth						
AIoT	28%	142%	72%	137%	85%	72%
Biometrics	-7%	18%	69%	0%	0%	0%
Industrial 3D scanning	16%	-13%	-2%	0%	0%	0%
3C electronics	-81%	-100%	-	-	-	-
Others	-20%	-11%	40%	40%	30%	20%

Source: Company data, J.P. Morgan estimates

Gross margin expansion is supported by high barriers to entry, favorable mix, and supply chain standardization

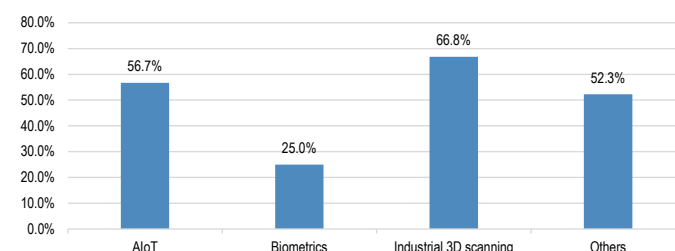
Orbbec's GPM has remained above 40% since 2019, and we believe this level is sustainable given the company's technology moat and established supply chain, including TSMC for foundry. New entrants face a steep learning curve in algorithm, chip design, and market-specific customization. We expect GPM to expand toward c.47% during 2026–2028E, driven by a favorable product mix shift toward AIoT. Humanoid robot and physical AI applications are likely to remain margin accretive in the near term, given the current demand-supply dynamics. We estimate AIoT segment GPM to stay above 50% during the forecast period, though we model a c.2ppts Y/Y decline due to capex expansion, while remaining cautiously optimistic about the competitive landscape.

Figure 12: Gross profit and blended GPM forecasts (in Rmb MM)



Source: Company data, J.P. Morgan estimates

Figure 13: Segmental GPM in comparison (2025A)

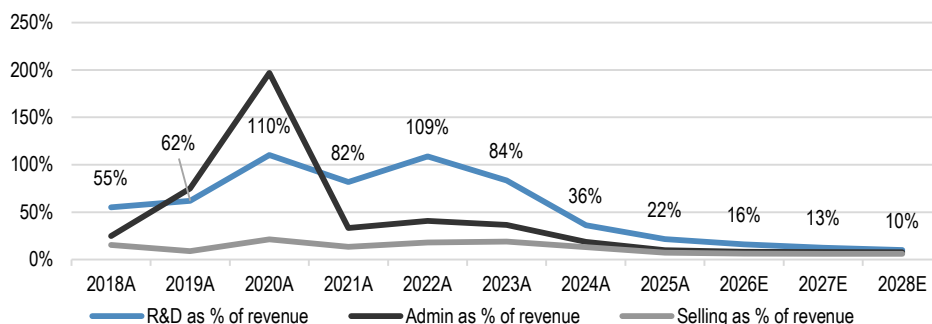


Source: Company data

Orbbec's disciplined cost structure and scale benefits are driving operating leverage and margin improvement

R&D expenses as a percentage of revenue have declined from a peak of 110% in 2020 to 22% in 2025, and we expect this ratio to fall further to c.10% by 2028E as revenue growth outpaces expense increases. The absolute R&D spend is projected to expand by c.Rmb60MM p.a. to support team expansion, compensation, and new product projects. Selling expenses as a percentage of revenue have dropped from c.20% in 2020 to 7% in 2025, and we expect further improvement to 6% by 2028E, reflecting scale benefits and the B2B nature of the business. Admin expenses are also modeled to decline as a percentage of revenue, supporting further operating leverage.

Figure 14: SG&A expenses as % of revenue trend forecasts

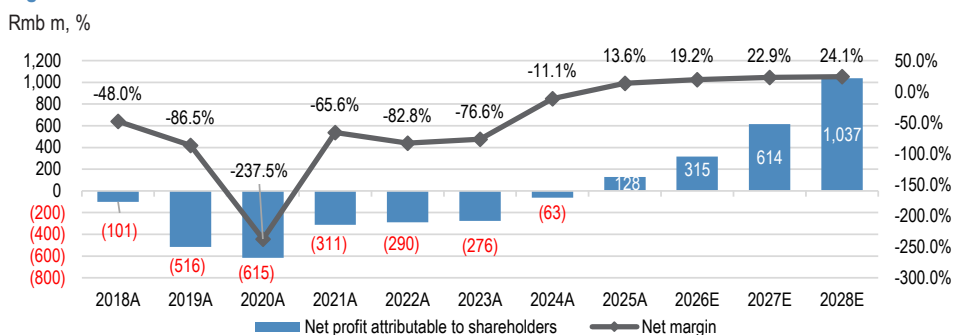


Source: Company data, J.P. Morgan estimates

Net margin and profit growth are accelerating as Orbbec leverages its supply chain and ecosystem position

Orbbec achieved a turnaround in 2025, recording a NPM of 14%. In management's blue-sky scenario, NPM could reach 30–40% within three years, leveraging the company's well-established supply chain and ecosystem position in the AIoT market. We project NPM to reach 24% by 2028E, with c.100% earnings growth CAGR over the forecast period.

Figure 15: NPAT and NPM trend forecasts



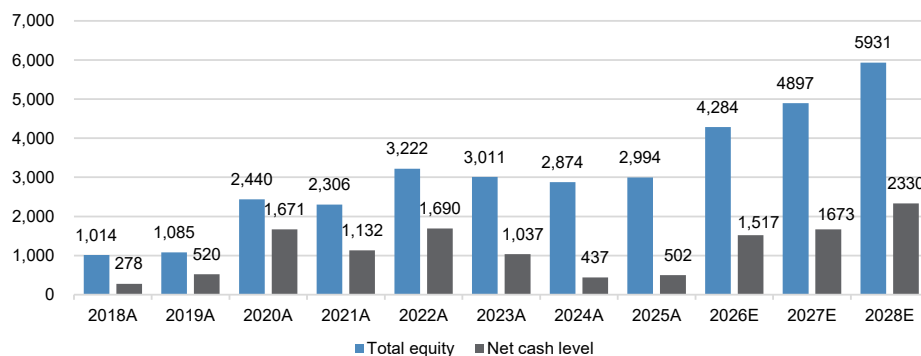
Source: Company data, J.P. Morgan estimates

Balance Sheet & CF Statement

Balance sheet strength and cash flow discipline support self-financing growth and capacity expansion

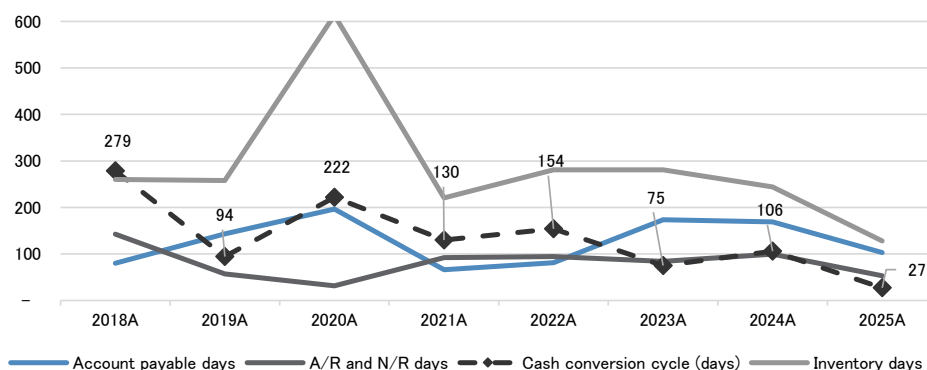
In 2025, Orbbec achieved a turnaround in Op-CF, generating Rmb83MM in operating cash inflow and an Op-CF/PBT ratio of 70%, underscoring its strong value chain position and pricing power. The cash conversion cycle improved to 27 days in 2025, down from a peak of 222 days, reflecting 128 days of inventory turnover, 53 days of receivables, and 103 days of payables. We expect this healthy cycle to be sustainable, given the short order-to-delivery cycle and semi-standard product formulation. As of end-2025, Orbbec had Rmb630MM in cash and equivalents. With anticipated sustainable Op-CF generation, we expect the company to maintain a self-financing virtuous cycle with minimal funding constraints. Management guided c.Rmb800MM capex for current capacity expansion, supporting a total output value of Rmb10B and well covering the modeled demand growth curve into 2030E.

Figure 16: Cash reserve and net debt in Rmb MM



Source: Company data, J.P. Morgan estimates

Figure 17: Working capital management



Source: Company data

Appendix: Key 3D vision technologies for humanoid robotics

This appendix summarizes the main 3D vision technologies relevant to humanoid robotics, with a focus on their operating principles, use cases, and limitations.

Structured light

Structured light projects a known pattern onto an object and calculates 3D shape by analyzing the deformation of the pattern. This method enables highly accurate, close-range 3D data capture (typically within 3–5 meters), making it well-suited for tasks requiring precision, such as manipulation and in-hand sensing. Orbbec's binocular structured light is currently the most adopted machine vision in humanoid robot applications. However, structured light struggles in poor lighting, with transparent or dark objects, and is not cost-effective for long-range navigation.

Binocular stereo vision

Stereo vision uses two cameras to recover depth by triangulating the disparity between images. This approach excels at real-time scene understanding and robust, close-range manipulation, such as wrist-mounted sensors for humanoid robots. It is effective for both dynamic and static scenes but can struggle with low-light or low-texture environments.

Monocular vision

Monocular vision estimates depth from a single camera, either by object size or by comparing frames over time (structure-from-motion). While it offers low hardware cost and rich visual detail, it lacks direct depth measurement and relies on advanced algorithms to bridge the gap. Monocular vision is best for data collection, indoor navigation, and scenarios where semantic understanding is key, but is less reliable for precise spatial tasks.

Time-of-flight (ToF)

ToF measures distance by timing how long it takes for emitted light to reflect back from an object. Direct ToF (dToF) is suitable for long-range navigation and outdoor use, while indirect ToF (iToF) is lower cost and often used for obstacle avoidance. iToF +RGB solutions are popular for world model development, enabling real-time 3D perception and active interaction. ToF's main drawbacks are cost (for dToF) and lower accuracy or multipath interference (for iToF).

LiDAR

LiDAR uses laser beams to scan the environment and generate a point cloud, making it ideal for SLAM and long-range navigation. For bipedal humanoids, LiDAR can affect stability and balance, so OEMs often use MEMS or OPA LiDAR to minimize impact. LiDAR is less suited for close-range manipulation and can be cost-prohibitive for some applications.

Industrial 3D measurement

Industrial 3D measurement combines triangulation and ToF, using laser scanning, structured light, or photogrammetry to create detailed CAD data for inspection and metrology.

Table 13: 3D machine vision technology comparison in summary

Technology	How it works	Typical measurement distance	Typical measurement accuracy	Strengths	Weaknesses	Output data type
Structured light	Projects an IR pattern and estimates depth from pattern deformation	Short range: ~0.2–3 m, sometimes up to ~5 m indoors	High at short range; often millimeter to centimeter level	Excellent close-range 3D shape capture; good for faces, hands, objects, and manipulation	Poor performance in strong sunlight; limited range; possible interference between devices	Dense depth map, RGB-D image, point cloud, surface/mesh reconstruction
iToF — indirect Time of Flight	Measures phase shift of modulated light	Short to mid range: ~0.2–10 m, sometimes ~15 m	Moderate to good; usually centimeter-level	Compact, fast, dense depth output, good indoor performance	Multipath interference; lower precision at long range; sunlight sensitivity	Dense depth map, amplitude/IR image, RGB-D image, point cloud
dToF — direct Time of Flight	Measures actual photon travel time	Mid to long range: ~0.5–50 m+, LiDAR can go much farther	Good at longer range; centimeter-level or better depending on sensor	Strong long-range performance; better outdoor robustness; useful for mapping and navigation	More expensive/complex; some modules have lower spatial resolution; higher power/integration cost	Sparse or semi-dense point cloud, range image, depth map, 3D occupancy data
Binocular stereo vision	Uses two cameras and triangulation, similar to human depth perception	Mid range: ~0.3–20 m depending on baseline and resolution	Good when texture and calibration are strong; centimeter-level at close/mid range, degrades with distance	Passive sensing; low cost; provides RGB plus geometric depth cues; aligns well with AI vision pipelines	Struggles with textureless, reflective, transparent, or low-light scenes; requires calibration	Left/right RGB images, disparity map, depth map, RGB-D image, point cloud
Monocular vision	Estimates depth from a single camera using geometry, SLAM, or AI models	Flexible range, from close to long distance, but scale can be ambiguous	Lowest absolute metric accuracy unless aided by IMU, SLAM, known scale, or learned priors	Cheapest and simplest; rich semantic input; highly compatible with VLM/VLA models	Depth is inferred rather than directly measured; weak absolute scale and precision	RGB image, estimated depth map, semantic segmentation, object detections, 2D/3D scene representation

Source: J.P. Morgan.

Investment Thesis, Valuation and Risks

Orbbec - A *(Overweight; Price Target: Rmb150.00)*

Investment Thesis

We are Overweight on Orbbec, a leading 3D vision solution provider based in Shenzhen, China. Orbbec is strategically positioned as a platform leader in 3D machine vision, a critical enabler for the emerging embodied AI era. The company has an estimated 95% market share (according to management) in China's humanoid robot vision hardware segment, supported by its full-stack technology platform. Orbbec's solutions are embedded in major robotics ecosystems, with partnerships spanning NVIDIA, Microsoft, and leading domestic players.

Valuation

Our December 2027 price target is Rmb 150 based on an FY27E P/E multiple of 23x. We believe this premium is justified by Orbbec's dominant 95% market share in humanoid robot machine vision and its strong exposure to the sector's volume ramp.

Risks to Rating and Price Target

Key downside risks include: 1) Slower than expected market growth pace, notably the key drivers including humanoid robot/embodied AI applications and 3D printing applications; 2) increasing competition diluting the profitability; 3) potential shifts in vision hardware standards towards lower ASP categories.

Orbbec - A: Summary of Financials

Income Statement						Cash Flow Statement					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	564	941	1,646	2,684	4,298	Cash flow from operating activities	(86)	83	277	542	937
COGS	(329)	(529)	(877)	(1,406)	(2,266)	o/w Depreciation & amortization	71	50	55	67	79
Gross profit	236	412	769	1,278	2,033	o/w Changes in working capital	(82)	(71)	(76)	(119)	(151)
SG&A	(383)	(364)	(505)	(713)	(1,022)	Cash flow from investing activities	(594)	15	(234)	(384)	(277)
Adj. EBITDA	(32)	163	363	693	1,178	o/w Capital expenditure	(203)	(63)	(252)	(404)	(304)
D&A	(71)	(50)	(55)	(67)	(79)	as % of sales	35.9%	6.7%	15.3%	15.1%	7.1%
Adj. EBIT	(102)	113	308	627	1,099	Cash flow from financing activities	39	(78)	983	(2)	(2)
Net Interest	36	8	16	18	25	o/w Dividends paid	-	-	-	-	-
Adj. PBT	(65)	119	324	645	1,125	o/w Shares issued/(repurchased)	0	14	975	0	0
Tax	3	9	(10)	(32)	(90)	o/w Net debt issued/(repaid)	40	(22)	11	0	0
Minority Interest	(0)	0	1	1	2	Net change in cash	(640)	18	1,026	156	657
Adj. Net Income	(63)	128	315	614	1,037	Adj. Free cash flow to firm	(326)	13	9	122	610
Reported EPS	(0.16)	0.32	0.77	1.49	2.52	y/y Growth	(7.7%)	(103.9%)	(25.4%)	1196.4%	401.5%
Adj. EPS	(0.16)	0.32	0.77	1.49	2.52						
DPS	-	-	-	-	-						
Payout ratio	-	-	-	-	-						
Shares outstanding	400	401	411	411	411						
Balance Sheet						Ratio Analysis					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	606	630	1,656	1,811	2,469	Gross margin	41.8%	43.8%	46.7%	47.6%	47.3%
Accounts receivable	154	137	217	354	568	EBITDA margin	(5.6%)	17.4%	22.0%	25.8%	27.4%
Inventories	220	186	320	512	826	EBIT margin	(18.1%)	12.1%	18.7%	23.4%	25.6%
Other current assets	449	507	524	549	588	Net profit margin	(11.1%)	13.6%	19.2%	22.9%	24.1%
Current assets	1,429	1,459	2,717	3,227	4,450	ROE	(2.1%)	4.4%	8.7%	13.4%	19.2%
PP&E	338	471	532	878	1,111	ROA	(1.9%)	3.8%	7.6%	11.6%	16.1%
LT investments	17	18	19	20	21	ROCE	(3.4%)	3.4%	7.9%	12.6%	18.3%
Other non current assets	1,545	1,464	1,601	1,591	1,584	SG&A/Sales	67.8%	38.7%	30.7%	26.6%	23.8%
Total assets	3,329	3,413	4,868	5,717	7,166	Net debt/Equity	NM	NM	NM	NM	NM
Short term borrowings	170	128	138	138	138	Net debt/EBITDA	13.8	NM	NM	NM	NM
Payables	152	150	248	397	672	Sales/Assets (x)	0.2	0.3	0.4	0.5	0.7
Other short term liabilities	81	99	156	243	384	Assets/Equity (x)	1.1	1.2	1.1	1.2	1.2
Current liabilities	403	377	543	779	1,193	Interest cover (x)	0.9	NM	NM	NM	NM
Long-term debt	0	0	0	0	0	Operating leverage	(114.2%)	(316.5%)	228.5%	164.6%	125.2%
Other long term liabilities	51	42	42	42	42	Tax rate	(3.9%)	7.4%	3.0%	5.0%	8.0%
Total liabilities	455	419	585	820	1,235	Revenue y/y Growth	56.8%	66.7%	75.0%	63.0%	60.2%
Shareholders' equity	2,867	2,977	4,267	4,881	5,918	EBITDA y/y Growth	(85.0%)	(616.3%)	122.3%	91.0%	69.8%
Minority interests	7	17	17	15	13	EPS y/y Growth	(77.2%)	(302.8%)	140.5%	94.8%	68.8%
Total liabilities & equity	3,329	3,413	4,868	5,717	7,166						
BVPS	-	-	-	-	-						
y/y Growth	-	-	-	-	-						
Net debt/(cash)	(437)	(502)	(1,517)	(1,673)	(2,330)						

Source: Company reports and J.P. Morgan estimates.

Note: Rmb in millions (except per-share data). Fiscal year ends Dec. o/w - out of which

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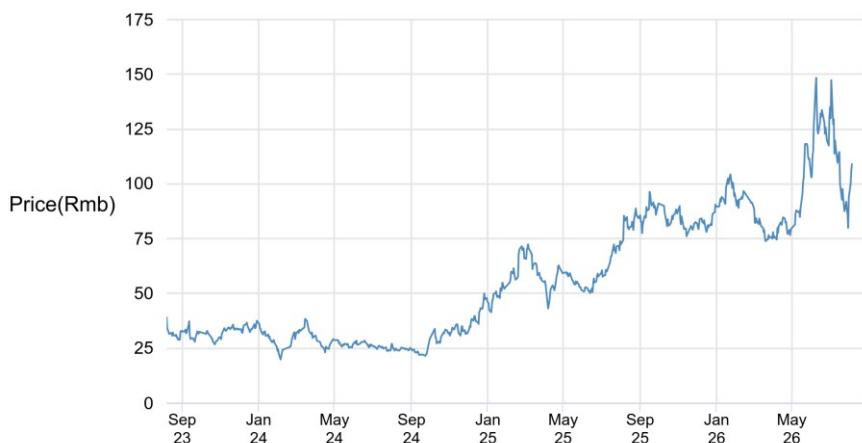
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Orbbec - A (688322.SS, 688322 CH) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.

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